# Nathan Espejo

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## Education

**Western University** — London, ON
*Bachelor of Engineering Science in Software Engineering, Dean's Honour List (2024, 2025)* — *Aug 2024 - May 2028*

## Experience

**VP of Data/ML (promoted from Software Developer)** — Sep 2025 - Aug 2026
*3D Western* — *London, ON*

- Co-led two data developers through Jira tickets and PR review on the print-approval dataset and benchmark.
- Built a labelled dataset from 4,620 scraped 3D models, rendered in Blender with labels checked by a VLM.
- Benchmarked small vision models as a binary safe-or-flag gate: 84.2% best agreement with VLM-checked labels.

**Software Engineering Intern** — May 2026 - Aug 2026
*TeraGo* — *Markham, ON*

- Built outage ticketing for non-standard provider emails: a local LLM (Ollama) extracts the key details and opens Jira issues through Microsoft Graph; works end to end, not yet deployed.
- Wrote Channel Finder, a Python tool that finds clear RF channels in MapInfo data, validated on 760 test cases.
- Deployed a Copilot Studio agent that answers engineers' RF questions from SharePoint documents.
- Proposed a self-hosted LLM platform (LibreChat with SharePoint and Jira) to cut Copilot licensing costs.

## Projects

**Campus** | *Python, FastAPI, SQLite, React, TypeScript, Docker* — Aug 2026 - Present

- Built a self-hosted AI study assistant that syncs my LMS courses into SQLite with Playwright and runs on any OpenAI-compatible endpoint, local or hosted.
- Designed hybrid retrieval over course files with embedding search, cross-encoder reranking and a lexical SQL prefilter. Fixed exact phrases that scored below the semantic cutoff.
- Wrote a 20-tool agent harness with an audit log for every AI write and a bounded tool loop, with 308 tests in CI.
- Use it across six courses for deadline lookups, studying from course slides, and reviewing drafts against briefs.

**Homelab & NixOS** | *NixOS, Hermes Agent, Docker, Tailscale* — May 2026 - Present

- Defined a laptop and home server in one git-tracked NixOS config, with 27 hardened containers behind Tailscale.
- Run one Hermes Agent profile per host that makes changes as config edits, across 472 sessions since June 2026.
- Gated agent changes behind my approval and a nixos-safe-rebuild guard, with rollback to the last good build.
- Grew a git-versioned library of 191 agent skills run on low-cost models picked per task, pruned weekly by a curator that keeps backups of removed skills.

**AWARE** (Hack the 6ix, solo) | *Python, YOLOv8, YAMNet, ONNX, Qualcomm QNN* — Jul 2026

- Built a home-awareness device running YOLOv8 and YAMNet on an $80 Arduino UNO Q, no cloud inference.
- Used an on-device LLM to compile commands like "when glass breaks after 10pm alert me" into deterministic rules, checked every 500 ms with no LLM in the loop.

## Hackathon Awards

**8 awards in 22 hackathons**: NSBEHacks 3rd place, 1st for Cohere (Careerly) | Hack Western Best AI Application with Cloudflare (Auralis) | TerraHacks Best Use of MongoDB (Lumen) | Spark 2026 3rd (CareBridge, contributor)

## Technical Skills

**AI**: LLM agents, tool calling, RAG, embeddings and reranking, MCP, VLMs, model evaluation, llama.cpp, Ollama, PyTorch, YOLOv8, ONNX
**Languages**: Python, TypeScript, C#, C++, SQL, Bash
**Web & Infra**: React, FastAPI, Next.js, NixOS, Linux, Docker, Tailscale, Caddy, Git, GitHub Actions
**Tools**: Hermes Agent, Microsoft Graph, Copilot Studio, Jira, SharePoint, Blender